



HOSPITAL-READY OPEN CLINICAL AI
PLATFORM · WHITEPAPER

Willow Health AI Suite

From eight open-source medical AI projects to one governed clinical AI stack — a productization blueprint covering architecture, compliance, deployment, ROI and a 90-day pilot path.

**Prepared for Clinical Operations & the AI Steering
Committee**

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WILLOW HEALTH AI · WHITEPAPER SERIES

Table of Contents

Executive Summary	2
The Case for an Open Clinical AI Stack	3
Platform Architecture: The Willow Suite	4
Source Projects: Deep-Dive and Improvisation Plan	6
3.1 MedGemma 1.5 (Google) - the Writer engine	7
3.2 BioMistral-7B (EU consortium) - the Sage engine	7
3.3 Meditron (EPFL) - the audited training pipeline	8
3.4 BioGPT (Microsoft) - the Pulse miner	8
3.5 medspaCy (JHU / Vanderbilt / NIH) - the assertion engine	9
3.6 scispaCy (AllenAI) - the UMLS coder	9
3.7 OpenBioLLM-8B (Saama / aaditya) - the Triage engine	10
3.8 Clinical-Longformer and Clinical-T5 (Stanford AIMI) - the Charts encoders	10
Product Lines for Clinical Operations	11
Compliance and Governance Framework	13
5.1 HIPAA: PHI routes and the de-identification gate	14
5.2 EU AI Act: high-risk obligations on a deadline	14
5.3 FDA: the SaMD boundary and the PCCP strategy	15
5.4 Governance structure and kill criteria	15
Deployment and Infrastructure	16
EHR Integration via HL7 FHIR	17
ROI and Total Cost of Ownership	18
Implementation Roadmap	19
Risk Register and Validation Protocol	20
10.1 Validation protocol	21
10.2 Kill criteria	21
References	22

Executive Summary

Clinical operations teams are spending an ever-growing share of their day on documentation rather than on patients. Peer-reviewed evidence now shows the scale of both the problem and the opportunity: clinician burnout fell from 51.9 percent to 38.8 percent within 30 days of introducing ambient AI documentation support (JAMA, 2025), and controlled studies consistently report 20 to 30 percent reductions in documentation time. Commercial platforms that deliver these gains exist, but they price at roughly 2,500 to 7,200 US dollars per clinician per year, lock hospitals into closed models, and move patient data through vendor infrastructure. At the same time, the open-source ecosystem has matured dramatically: Google's MedGemma 1.5, BioMistral-7B, EPFL's Meditron pipeline, and Stanford AIMI's Clinical-Longformer and Clinical-T5 encoders now operate at or near clinical-benchmark parity with proprietary systems.

What open source has never shipped, however, is the wrapper that makes any of it hospital-grade: de-identification, guardrails, audit trails, validation protocols, and a governance structure that a compliance officer would sign off on. That gap is exactly where this whitepaper positions the Willow Health AI Suite. Rather than deploying eight research projects as eight disconnected tools, Willow consolidates them into a single governed platform with four layers: a Trust Fabric that wraps every model call, an Intelligence layer that serves fine-tuned open models, a Data and Integration layer built on HL7 FHIR, and an Experience layer of six clinician-facing product lines. Every one of the eight source projects is retained, upgraded, and hardened in the process - none is discarded.

51.9% to 38.8%

Clinician burnout within 30 days of ambient AI documentation support
(JAMA, 2025)

The economics support the investment. A self-hosted Willow deployment is estimated at 900 to 1,400 US dollars per clinician per year at 100-clinician scale, including hardware amortization and a three-person support team - roughly one third to one fifth of typical commercial scribe contracts - with payback on hardware in 12 to 20 months. The deepest chapter of this document is deliberately the compliance chapter, because that is the hardest part: HIPAA de-identification, EU AI Act high-risk obligations that become fully enforceable in August 2026, and FDA software-as-a-medical-device boundary analysis each demand engineering and governance commitments that no raw model download provides. The recommended next step is a 90-day, two-department pilot of Willow Writer under the governance framework described in Chapter 9, with explicit go and no-go gates agreed in advance.

The Case for an Open Clinical AI Stack

The documentation burden in hospitals is no longer a soft complaint; it is a measured operational cost. Time-motion studies repeatedly show clinicians spending one to two hours on EHR and desk work for every hour of direct patient contact, with the EHR inbox alone consuming over an hour per day in primary care. The consequence is well quantified: burnout rates above 50 percent in several specialties, rising turnover, and attrition costs that hospitals absorb in recruitment and locum cover. Ambient and generative AI documentation tools are the first intervention class with randomized and quasi-experimental evidence of relief at scale, which is why health systems have raced to adopt them.

The commercial market, however, prices that relief as a perpetual per-seat subscription with no path to ownership. Analyst estimates for 2026 place enterprise contracts for tools such as Abridge and Nuance Dragon Copilot between 2,500 and 7,200 US dollars per clinician per year; a 100-clinician department therefore commits 250,000 to 700,000 US dollars annually, forever, with the model weights, prompts, and evaluation data all owned by the vendor. Closed systems also complicate privacy review: PHI leaves the hospital boundary, and the hospital cannot independently verify what the model does with it. For an institution that intends to build durable internal capability - rather than rent it - the open-weight ecosystem has crossed the quality threshold.

Table 1 summarizes the eight projects this platform is built from. Two facts stand out. First, coverage is remarkably complementary: multimodal medical foundation models (MedGemma), domain-pretrained reasoners (BioMistral, Meditron), clinical assertion engines (medspaCy), entity-linking toolkits (scispaCy), long-document encoders (Clinical-Longformer, Clinical-T5), and narrow high-precision miners (BioGPT) together cover nearly every clinical-language workload a hospital has. Second, licenses are workable for hospital use, with two caveats - the Gemma terms attach use restrictions to derivatives, and PhysioNet-credentialed models inherit a data use agreement - both manageable and addressed in Chapter 5. What none of them ships is hospital readiness itself, which is the productization work described in the rest of this document.

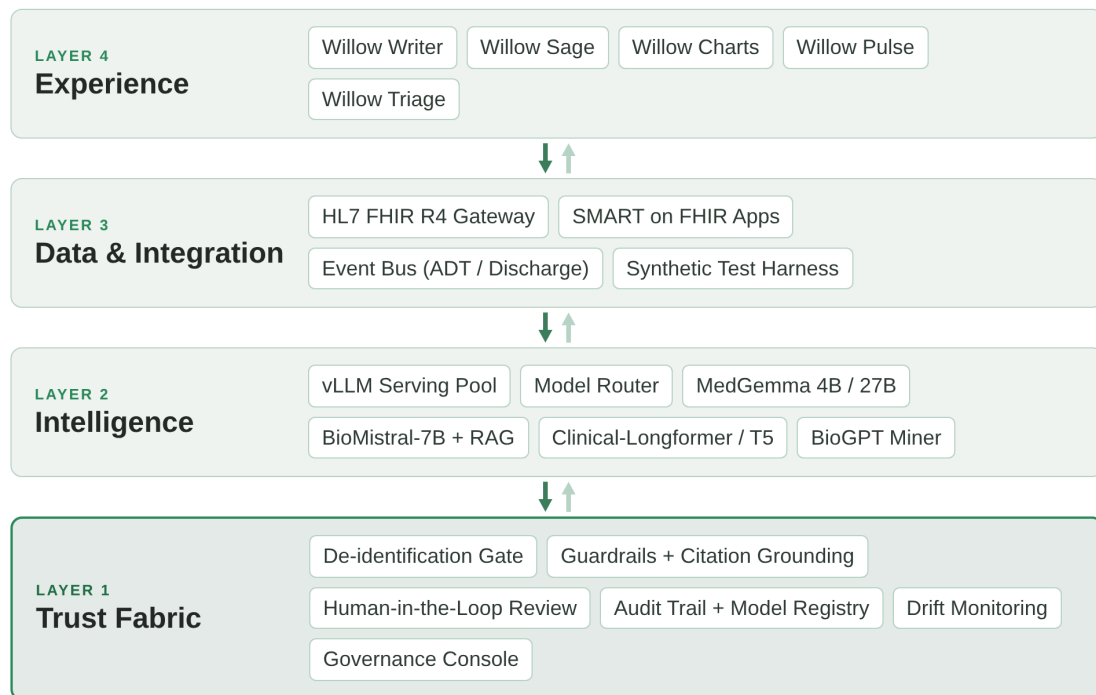
Project	Origin	Class	License	Role in Willow
MedGemma 1.5 (4B / 27B)	Google	Multimodal medical LLM	Gemma Terms of Use	Primary writer / multimodal engine
BioMistral-7B	EU consortium	Domain-pretrained LLM	Apache-2.0	Guideline QA reasoner + RAG
Meditron (7B / 70B)	EPFL	Clinical LLM pipeline	Llama-2 Community	Audited training pipeline
BioGPT	Microsoft	Biomedical transformer	MIT	Relation mining / evidence digest
medspaCy	JHU / Vanderbilt / NIH	Rule-based clinical NLP	MIT	Assertion + section engine
scispaCy	AllenAI	Biomedical NLP models	Apache-2.0 (code)	NER + UMLS entity linking
OpenBioLLM-8B	Saama / aaditya	Fine-tuned medical LLM	Llama-3 Community	Triage reasoning engine
Clinical-Longformer / Clinical-T5	Stanford AIMI	MIMIC-pretrained encoders	PhysioNet-credentialed	Risk heads on long notes

Table 1. The eight source projects at a glance

The strategic logic is therefore not open versus commercial on quality alone - it is ownership versus rental, transparency versus black box, and fixed cost versus perpetual per-seat fees. Open weights let the hospital fine-tune on its own note style, pin versions in a registry, and pass audits with full artifact traceability. The remainder of this whitepaper describes how to convert that potential into a governed, production-grade clinical platform.

Platform Architecture: The Willow Suite

Willow Health AI Suite is a four-layer platform in which every clinical capability shares one governance core. The design principle is governance-first: rather than bolting safety onto individual apps, the platform inserts a Trust Fabric between the models and everything else, so that no protected health information ever reaches a model un-de-identified and no generated output reaches a patient record without human review. This inverts the usual open-source failure mode, where impressive demos fail procurement precisely because the compliance wrapper was never engineered. Figure 1 shows the layers and their components.



Every model call is wrapped by the Trust Fabric: no PHI reaches a model un-de-identified, no output reaches the chart un-reviewed.

Figure 1. Willow Suite four-layer platform architecture

The Intelligence layer runs a shared vLLM serving pool behind a model router, which is what makes the platform economical: a 7B model on a single 24 GB GPU serves Sage and Triage traffic, MedGemma 4B handles Writer drafts, and the encoder fleet runs on modest hardware, all scaled independently per department workload. The Data and Integration layer exposes everything through an HL7 FHIR R4 gateway with SMART on FHIR launch contexts, so products appear inside the existing EHR workflow instead of as a separate tab clinicians must remember to open. The Experience layer packages capabilities into the six product lines detailed in Chapter 4, each with its own KPI set and its own go and no-go criteria.

Layer	Components	Function
4 Experience	Writer, Sage, Charts, Pulse, Triage apps	Clinician-facing workflow entry points with per-product KPIs
3 Data / Integration	FHIR R4 gateway, SMART on FHIR, event bus, synthetic harness	EHR context in, signed output back; ADT and discharge triggers
2 Intelligence	vLLM pool, model router, fine-tuned MedGemma, BioMistral+RAG, Clinical-Longformer/T5, BioGPT	Serving, routing and tuning of all open models
1 Trust Fabric	De-ID gate, guardrails and citation grounding, human-in-the-loop, audit trail and model registry, drift monitoring, governance console	Safety, compliance and evidence for every model call

Table 2. Layer responsibilities and component inventory

Two architectural decisions deserve emphasis. First, the model router means models are swappable, not load-bearing: if a license changes or a better open model appears, the platform upgrades without re-platforming, which neutralizes the main long-term risk of open-source dependence. Second, the Trust Fabric is itself a product (Willow Sentinel) with a user interface, dashboards, and an owner - governance that is visible and staffed, rather than a policy document in a drawer. This is the single most important differentiator between a pile of checkpoints on Hugging Face and a system a hospital can actually run.

Source Projects: Deep-Dive and Improvisation Plan

Each of the eight projects was reviewed individually: what it verifiably is today, what license governs it, where it falls short of hospital use, and the specific improvisation Willow applies. The pattern is consistent - keep the trained knowledge in the weights, replace the research scaffolding with production infrastructure, and harden with the Trust Fabric. None of the eight is used as-is; none is discarded either.

3.1 MedGemma 1.5 (Google) - the Writer engine

MedGemma 1.5 is Google's open medical vision-language family built on Gemma 3, shipped in a 4B multimodal variant with a SigLIP image encoder pre-trained on de-identified medical data - chest X-rays, CT and MRI volumes, histopathology and dermatology imaging - and a 27B text-heavy variant. The 1.5 release improved MedQA accuracy by 5 percent and EHRQA by 22 percent over the original, and Google explicitly expects developers to fine-tune it for institutional use cases. The license is the Gemma Terms of Use: free for research and commercial use, but with use restrictions that propagate to derivative models, which Chapter 5 handles through registry pinning and legal review. Willow keeps MedGemma as the multimodal core of Willow Writer and, in the 27B variant, as the escalation model for complex multimodal review.

Keep: multimodal medical foundation capability; benchmark trajectory across versions.

Add: supervised fine-tuning on hospital discharge summaries and after-visit notes with clinician-graded outcomes.

Harden: structured-output constraints, citation grounding to chart context, full Gemma-terms compliance tracking in the model registry.

3.2 BioMistral-7B (EU consortium) - the Sage engine

BioMistral-7B is Mistral-7B-Instruct further pre-trained on the complete PubMed Central corpus and released under Apache-2.0, the cleanest license in this portfolio. It is competitive across ten medical question-answering benchmarks at 7B scale, has spawned follow-up work (BioMistral-NLU) generalizing its medical natural-language understanding, and is EU-hostable - a meaningful property for European data-residency requirements. Its weakness is exactly what raw pre-training never provides: it can sound authoritative while being wrong. Willow therefore deploys it exclusively behind retrieval-augmented generation over the hospital's own guideline library - NICE, WHO and institutional protocols - with every claim citation-grounded and answers that cannot cite a retrieved passage are refused rather than guessed.

Keep: Apache-2.0 weights; PubMed Central domain knowledge; EU hosting option.

Add: MEGA-RAG-style citation-grounded retrieval over local guidelines with refusal-on-no-evidence behavior.

Harden: guardrail layer per Nature npj Digital Medicine guidance; monthly benchmark regression suite.

3.3 Meditron (EPFL) - the audited training pipeline

Meditron is EPFL's suite of medical LLMs adapted from Llama-2, pre-trained on clinical guidelines (WHO, UK NHS) plus PubMed abstracts and MIMIC-III/IV notes, and it was the strongest open medical model at its 2023 release. Its most valuable contribution to Willow in 2026 is not the weights but the methodology: the Fully Open Meditron pipeline (May 2026) publishes the first fully auditable LLM-CDSS recipe - a clinician-audited training corpus and a reproducible data pipeline - which is precisely the evidence trail a hospital must produce under EU AI Act technical-documentation obligations. Willow adopts the audited pipeline as the standard for all institutional fine-tuning, rather than treating training as an artisanal, undocumented activity.

Keep: guideline-grounded training corpus methodology; clinician-audit workflow.

Add: institution-specific small-model training for note style and local protocol vocabulary.

Harden: every training run versioned in the registry with corpus hashes and reviewer sign-off.

3.4 BioGPT (Microsoft) - the Pulse miner

BioGPT is Microsoft's generative transformer for biomedical text, MIT-licensed including the pre-trained weights, with state-of-the-art results on BC5CDR, KD-DTI and DDI relation extraction and 78.2 percent on PubMedQA at publication. It is narrow by modern LLM standards - 346M parameters, BART-style generation - and that is exactly why it survives the cut: precision tasks with no hallucination tolerance, where a small deterministic specialist beats a large generalist. Willow uses BioGPT inside Willow Pulse for drug-disease and gene-disease relation extraction over incoming literature, feeding the daily evidence digest, and as a candidate-labeler to pre-screen publications before any generative summarization happens.

Keep: MIT-licensed relation-extraction checkpoints; PubMedQA-class QA ability.

Add: department interest profiles that rank the daily PubMed/MEDLINE digest per research team.

Harden: extraction-only role in the pipeline; outputs always linked to source article identifiers.

3.5 medspaCy (JHU / Vanderbilt / NIH) - the assertion engine

medspaCy is the open clinical NLP toolkit behind a widely cited JAMA-format paper (Eyre et al., 2022), providing ConText-based assertion detection - negation, family history, uncertainty, experiencer and temporality - plus section detection and hybrid rule-and-statistical concept extraction. These are unglamorous capabilities with outsized safety value: a model that reads 'father had MI' as the patient having an MI is not a better model, it is a dangerous one. Willow embeds medspaCy in every pipeline that touches free text: Writer uses section detection to structure drafts, Charts uses ConText to qualify extracted findings, and the de-identification gate uses its dictionaries as a first-pass detector. Its MIT license and pure-Python deployment make it the lowest-risk component in the stack.

Keep: ConText assertion engine; section detection; hybrid extraction architecture.

Add: institutional ICD-10 and SNOMED coding dictionaries and local shorthand lists.

Harden: quarterly dictionary review with clinical informatics; assertion output logged for audit.

3.6 scispaCy (AllenAI) - the UMLS coder

scispaCy provides spaCy pipelines and models for biomedical and scientific text, including NER models trained on BC5CDR and related corpora, abbreviation detection, and - critically - entity linking into UMLS, MeSH and RxNorm knowledge bases. That linking step is what turns 'atrial fibrillation' into a coded, queryable, billable, decision-support-ready concept, and it is the bridge between free-text notes and the structured world the EHR and coding teams live in. Willow fine-tunes its NER on cardiology notes to auto-extract LVEF values, NYHA class and medication lists for Willow Charts, and runs the UMLS linker in batch mode to maintain a coded problem list alongside the clinician's own. Because scispaCy models are small and CPU-capable, this runs continuously and cheaply on the encoder fleet.

Keep: BC5CDR-class NER; UMLS/MeSH/RxNorm entity linking; abbreviation detection.

Add: cardiology-domain NER fine-tune (LVEF, NYHA, meds) with a gold-standard held-out set.

Harden: extraction confidence thresholds; ambiguous-link flags routed to human coder queue.

3.7 OpenBioLLM-8B (Saama / aaditya) - the Triage engine

OpenBioLLM-8B is a Llama-3-8B derivative tuned with domain-adaptive pre-training and preference optimization on medical data, averaging 72.5 percent across nine biomedical benchmarks - outperforming GPT-3.5-class and Meditron-70B at one tenth the size, with the 70B variant reaching 86.1 percent. Under the Llama-3 Community License (commercial use permitted below 700 million monthly active users), it gives Willow a strong medical reasoner that fits on a single 24 GB GPU. Willow uses it inside Willow Triage for structured symptom-intake reasoning and pre-consultation summaries - strictly as a decision-support aid with an RN review gate and deterministic escalation rules for red-flag symptoms, never as an autonomous decision maker. The multi-agent guardrail patterns from the 2026 clinical-safety literature wrap every triage turn.

Keep: 8B medical reasoning quality; single-GPU footprint; Llama-3 Community License.

Add: structured intake schema, red-flag escalation rules, telehealth front-end.

Harden: RN review gate; no-triage-without-human sign-off; conversation transcripts in the audit trail.

3.8 Clinical-Longformer and Clinical-T5 (Stanford AIMI) - the Charts encoders

Clinical-Longformer adapts Longformer's 4,096-token sparse attention to MIMIC-III notes and consistently outperforms ClinicalBERT-class models on long-document clinical tasks, while Clinical-T5 provides four span-corruption models pre-trained on the union of MIMIC-III and MIMIC-IV notes. A 2025 PLOS study demonstrates the practical pattern: Clinical-T5 embeddings combined with structured EHR data predict 30-day readmission with clinically usable accuracy. Both are hosted on PhysioNet, which means access is credentialed and governed by a MIMIC data use agreement - a compliance nuance the registry tracks explicitly. Willow fine-tunes them as the readmission and sepsis risk heads of Willow Charts, deployed in silent mode until Chapter 10's validation gates are passed.

Keep: 4,096-token long-note encoding; MIMIC-III/IV pre-training; proven readmission pattern.

Add: hospital-cohort fine-tuning heads; silent-mode evaluation against 90-day outcomes.

Harden: PhysioNet DUA tracking in the registry; drift monitoring on input note distributions.

Product Lines for Clinical Operations

Architecture only matters when it changes someone's Tuesday afternoon. The six Willow product lines are therefore described by workflow - who touches them, when, and what changes for the clinician - rather than by model. Each product ships with its own KPI set, its own review gate, and its own kill criteria; no product writes to the record without a named human owner approving the output. Table 3 maps every product to its engine, its EHR surface and its primary KPI.

Willow Writer - discharge and after-visit documentation

Writer monitors the discharge event on the ADT feed, assembles chart context through the FHIR gateway, and drafts a discharge summary or after-visit note in the hospital's own house style within 90 seconds of the trigger. The draft opens in the clinician's existing EHR context via SMART on FHIR with every factual claim linked to its source note; the clinician edits and signs, and the AI never auto-signs anything. Target outcomes are a 20 to 30 percent reduction in documentation time per discharge and same-day sign-off rates above 90 percent in pilot departments.

Willow Sage - guideline-grounded clinical Q&A;

Sage answers protocol and policy questions for clinicians and operations staff - dosing thresholds, isolation rules, referral pathways - by retrieving from the hospital's controlled document library and generating an answer in which every sentence carries a citation to a retrieved passage. When retrieval confidence is low, Sage refuses and routes the question to the owning committee instead of improvising; the refusal log is itself a governance signal showing which guideline gaps matter most to staff. The product KPI is simple: 100 percent of answers citation-grounded, zero uncited factual claims in weekly audits.

Willow Charts - note intelligence and risk flags

Charts runs the scispaCy and medspaCy pipelines continuously over new notes, maintaining coded extractions (LVEF, NYHA class, medication changes) and firing Clinical-Longformer/T5-based readmission and sepsis risk scores into a daily ward-level digest. It deliberately ships as a digest, not as interruptive alerts: the 2026 literature on AI-in-CDSS is blunt that unmanaged alerting produces fatigue and abandonment. After a 90-day silent-mode validation against actual outcomes, threshold-tuned flags are surfaced to charge nurses and the care-management team.

Willow Pulse - evidence digest and relation mining

Pulse ingests the daily PubMed/MEDLINE feed, runs BioGPT relation extraction to pre-screen articles against each department's registered interest profile, and produces a morning digest of ranked, linked summaries - two minutes of reading per specialty, not two hours of searching. Research teams subscribe to entity profiles (a drug, a pathway, a device class) and the digest keeps a visible provenance chain back to every source article. The KPI is engagement: subscriptions active and reviewed weekly by at least 60 percent of subscribing clinicians after month three.

Willow Triage - telehealth pre-consultation

Triage is a structured intake front-end for telehealth and nurse hotlines: OpenBioLLM-8B reasons over a structured symptom schema to produce a pre-consultation summary and a suggested urgency tier, which a registered nurse reviews before anything enters the record. Red-flag symptom patterns trigger deterministic escalation paths that bypass the model entirely - chest pain protocols, for example, are rule-based by design. The product's contract with the hospital is explicit: it increases telehealth throughput and consistency, and it never makes an autonomous clinical decision.

Willow Sentinel - the compliance console

Sentinel is the product the other five cannot live without: the operations console for the Trust Fabric, exposing the de-identification gate's sampling QA, the model registry with pinned weights and licenses, drift dashboards per model and per input stream, the full audit trail of every model call, and the incident-response workflow. It is designed for the compliance officer and the ML operations engineer to share one screen, which is rarer than it should be and is precisely what makes the rest of the platform defensible in an audit. Its KPI is coverage: 100 percent of model calls logged, every model version traceable to corpus, license and reviewer.

Product	Engine stack	EHR surface	Primary KPI
Writer	MedGemma 4B fine-tune + medspaCy structuring	SMART on FHIR panel + note write-back	Draft in under 90 s; 20-30% time saved
Sage	BioMistral-7B + citation RAG over guideline library	EHR-embedded search + web	100% citation-grounded answers
Charts	scispaCy + medspaCy + Clinical-Longformer/T5 heads	Ward digest + coded lists	Silent-mode AUROC gate passed
Pulse	BioGPT relation mining + interest profiles	Email / intranet digest	60% weekly clinician engagement
Triage	OpenBioLLM-8B + rule-based escalation	Telehealth front-end	RN sign-off on 100% of encounters
Sentinel	De-ID gate + registry + audit + drift monitors	Governance console	100% of calls logged and traceable

Table 3. Product-to-engine mapping and primary KPIs

Compliance and Governance Framework

Compliance is the hardest part of hospital AI - harder than model quality - because the failure modes are organizational, legal and reputational rather than statistical. An open model with 95 percent benchmark accuracy that leaks PHI once can end a CIO's career; a beautifully governed model at the same accuracy is a deployable asset. This chapter is therefore deliberately the deepest in the whitepaper, and it treats the three regimes that govern the platform - HIPAA, the EU AI Act and FDA device regulation - as engineering requirements with named owners, not as a legal appendix. Where a requirement is genuinely open to interpretation, the framework states the hospital's chosen position explicitly so it can be reviewed and signed, which is all any regulator actually asks of a deployer.

5.1 HIPAA: PHI routes and the de-identification gate

HIPAA permits four practical routes for running LLMs against protected health information: cloud model endpoints under a Business Associate Agreement, dedicated vendor capacity, self-hosted open weights, and fully on-premise GPUs. Willow uses the last two exclusively - self-hosted and on-premise - because they keep PHI inside the hospital security boundary, eliminate third-party BAAs for inference, and make every access auditable in one place. The de-identification gate applies HIPAA Safe Harbor logic (the 18 identifier categories per HHS guidance) using a Philter-class open toolkit plus medspaCy dictionaries, targeting above 95 percent recall with a human sampling program: one percent of de-identified traffic is double-read by trained staff, and any miss triggers re-tuning and a logged corrective action. Identified and de-identified zones are physically separated network segments, and re-identification attempts are both technically prevented and policy-prohibited.

5.2 EU AI Act: high-risk obligations on a deadline

The EU AI Act (Regulation 2024/1689) classifies AI medical devices as high-risk, and the core obligations for high-risk systems become fully enforceable in August 2026 - conformity assessment, technical documentation, human oversight, logging, data governance and risk management. Hospitals deploying in-scope systems are themselves on the hook as deployers, which makes waiting a strategic risk rather than a prudent delay. Willow's position is to treat the Act's requirements as the platform's minimum feature list, as Table 4 shows; the Fully Open Meditron audited-pipeline methodology gives every fine-tuned model a reproducible corpus and training record, which is the technical-documentation foundation the conformity assessment demands. Products whose regulatory posture is ambiguous are shipped in monitoring-only mode until legal review resolves them.

Obligation (enforceable Aug 2026)	Willow deliverable
Risk management system	Sentinel risk register per product; kill criteria; incident workflow with owners
Technical documentation	Meditron-style audited corpus records; registry entries per model version
Data governance	Documented provenance of all training data; de-ID gate for operational data
Human oversight	Human-in-the-loop review in every product; no autonomous write paths
Logging and record-keeping	Immutable audit trail of every model call, input hash and output disposition
Conformity assessment support	Per-product documentation pack maintained in Sentinel, reviewed quarterly

Table 4. EU AI Act high-risk obligations mapped to Willow deliverables

5.3 FDA: the SaMD boundary and the PCCP strategy

FDA regulation turns on whether a software function is a medical device - intended to diagnose, cure, mitigate, treat or prevent disease. Willow's boundary analysis (Table 5) classifies Writer, Sage and Pulse as clinical and administrative workflow tools that inform clinicians but do not drive device-level decisions; Charts risk scores and Triage sit closer to the line and are treated as candidate SaMD with a deliberately conservative posture. For anything that may cross into device territory, the platform adopts the FDA's Predetermined Change Control Plan approach (final guidance December 2024; draft lifecycle guidance January 2025): pre-declared update protocols, pre-specified performance thresholds and revalidation procedures, so model improvements ship inside an approved change envelope rather than as uncontrolled drift. Silent-mode-first deployment - models run and are measured for a full quarter before any human sees their output - is the standing rule for every candidate-SaMD component.

Product	SaMD posture	Rationale	Control posture
Writer	Not a device (intended)	Drafts text for clinician edit; no diagnostic claim	Workflow tool; human sign-off mandatory
Sage	Not a device (intended)	Retrieves and cites controlled guidelines	Citation grounding; refusal behavior
Pulse	Not a device (intended)	Literature digest; no patient-level output	Provenance links to source articles
Charts / Triage	Candidate SaMD	Patient-level risk and urgency outputs	Silent-mode validation; PCCP draft before live use

Table 5. FDA SaMD boundary analysis per product line

5.4 Governance structure and kill criteria

The AI steering committee is the governance instrument: a standing body chaired by the CMIO with subgroups for clinical validation, data governance and ML operations, meeting monthly with authority to pause any product. Every Willow product carries a validation file (benchmarks, silent-mode results, clinician review sampling rates), a named clinical owner, and pre-agreed kill criteria - conditions under which the product is switched off without a meeting: PHI incident at the gate, hallucination rate above threshold in weekly audit, AUROC below the silent-mode gate, or two consecutive drift alarms. Pre-agreed kill criteria are what make AI governance real; a committee that must debate each failure from scratch will always be too slow. Shadow use is treated as a governance defect with its own corrective action: all Willow apps are SSO-gated, and usage outside intended workflows is surfaced by Sentinel's audit analytics.

Deployment and Infrastructure

The reference deployment is fully on-premise: inference never leaves the hospital network, and the platform runs as containers on Kubernetes in four security zones - a DMZ for the FHIR gateway, an application zone for Willow services, a data zone for the de-identified store and vector indexes, and a restricted model zone where the GPU fleet lives. Serving uses vLLM with continuous batching; 4-bit quantization (GPTQ/AWQ) is the default for models above 8B so that 27B-class models fit a 48 GB card with acceptable latency. The encoder fleet (scispaCy, medspaCy, Longformer/T5 heads) is CPU-capable and runs continuously at low cost. Backups follow the 3-2-1 rule with daily configuration snapshots; the full stack is redeployable from the registry in under a day, which is also the disaster-recovery commitment.

Model class	GPU requirement	Serving mode	Typical throughput
7B (BioMistral, OpenBioLLM)	1 x 24 GB (L40S / A10)	vLLM, 4-bit	~120 tok/s
27B (MedGemma text)	1 x 48 GB or 2 x 24 GB	vLLM, 4-bit	~40 tok/s
Encoder fleet + risk heads	CPU or 1 x 16 GB	Batch queues	Continuous
Optional 70B (Meditron)	2-4 x 80 GB (A100 / H100)	vLLM, 4-bit	~25 tok/s

Table 6. GPU sizing by model class

Hospital	GPU fleet	Estimated hardware capex	Scope
200 beds	2 x L40S	USD 60,000 - 80,000	Writer + Sage + encoders
500 beds	4 x L40S + 1 x A100	USD 120,000 - 160,000	All six products
1,000 beds	8 x L40S + 2 x H100	USD 250,000 - 350,000	All products, multi-site ready

Table 7. Reference configurations by hospital size

Monitoring covers the serving layer and the statistical layer: GPU utilization, queue depth and token latency drive capacity decisions, while Sentinel's drift monitors watch input note distributions, output length and refusal rates per model, alerting on two consecutive alarms. Capacity is planned with headroom - the 500-bed configuration runs a target 60 percent utilization at peak - because a documentation copilot that queues at 08:00 on Mondays will be abandoned within a month, and abandonment is the most expensive failure mode of all. Model weights are pinned in an internal registry with corpus hashes and license records, which doubles as the EU AI Act technical-documentation substrate described in Chapter 5.

EHR Integration via HL7 FHIR

Integration is a workflow problem before it is a technical one, and the platform's rule is that clinicians never leave the EHR to use Willow. The FHIR R4 gateway exposes context reads (Patient, Encounter, DocumentReference, Condition, MedicationRequest) and SMART on FHIR launch contexts, so Writer opens as an EHR panel already scoped to the right patient and encounter. Event-driven triggers come off the ADT feed: discharge events invoke Writer, new note events feed Charts, and guideline publication events invalidate the relevant Sage retrieval chunks. All traffic is logged end-to-end by Sentinel, which is what makes the integration audit-ready rather than merely functional. Figure 2 shows the round trip.

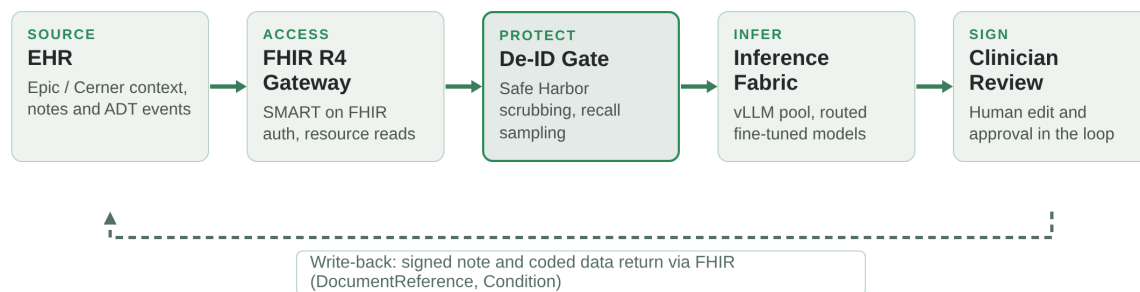


Figure 2. FHIR integration round trip: read, protect, infer, review, write back

Write-back is conservative by policy: Writer produces signed-note documents as DocumentReference resources with the human author as author and the model recorded in provenance; Charts contributes to a parallel coded layer rather than overwriting clinician-entered data; nothing ever writes to medication or order pathways. This preserves clinical accountability while still giving coding, quality and analytics teams the structured exhaust they need. Testing runs on Synthea synthetic patients end-to-end before any real chart is touched, and the integration rollout is phased by read scope: Phase 1 reads demographics and documents for Writer only; Phase 2 adds the ADT event bus and Labs for Charts; Phase 3 enables write-back beyond documents after two clean audit cycles. EHR-vendor specifics (Epic App Orchard vs SMART standards track) are an implementation detail the gateway isolates the products from.

ROI and Total Cost of Ownership

The financial model compares three-year total cost of ownership for a 100-clinician deployment. Commercial documentation AI prices at 2,500 to 7,200 US dollars per clinician per year in 2026 analyst estimates - 250,000 to 720,000 US dollars annually, recurring, with the estate owned by the vendor. Willow's cost structure is instead front-loaded: hardware (100,000 to 160,000 US dollars for the 500-bed class), a three-person run team (two MLOps engineers, one clinical informaticist, roughly 450,000 US dollars annually loaded), fine-tuning compute and evaluation overhead. At scale that lands at roughly 900 to 1,400 US dollars per clinician per year all-in - one third to one fifth of commercial pricing - and the hospital owns the weights, the data and the capability outright. Figure 3 visualizes the per-clinician contrast.

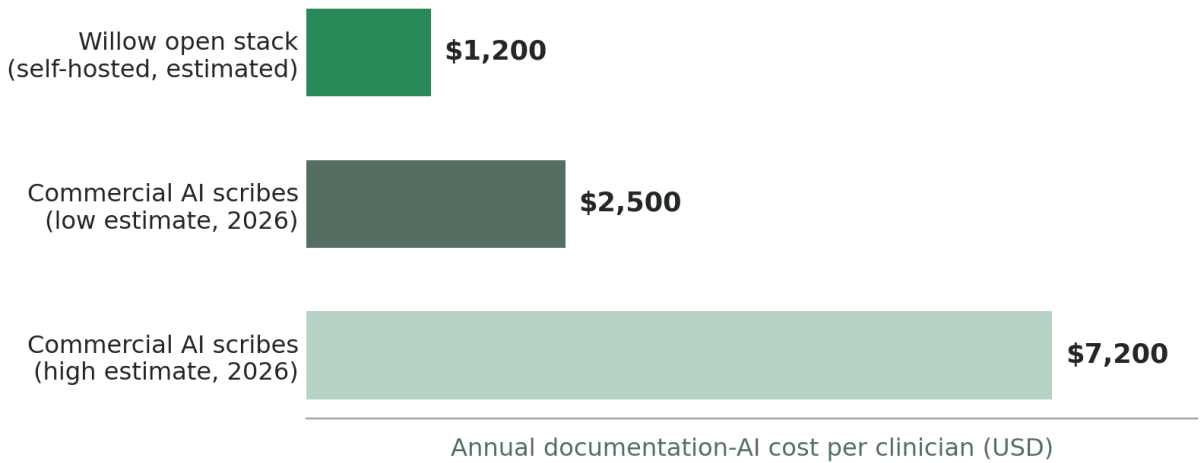


Figure 3. Annual documentation-AI cost per clinician: Willow open stack vs commercial scribes

Cost line	Willow (3 yr)	Commercial low (3 yr)	Commercial high (3 yr)
Software licenses / subscriptions	0	7,500	21,600
Hardware (amortized 3 yr)	40 - 55	included	included
Run team (3 FTE, 3 yr)	1,350	vendor-provided	vendor-provided
Fine-tuning + evaluation	90 - 120	n/a	n/a
Three-year total	1,480 - 1,525	7,500	21,600

Table 8. Three-year TCO sketch, 100-clinician deployment (USD thousands)

The benefit side compounds the case. At the JAMA-anchored 20 to 30 percent documentation-time reduction, a 100-clinician institution recovers roughly 8,000 to 12,000 clinician hours annually; valued conservatively at clinical-assistant rates, that alone covers the Willow run team. Burnout and retention effects are harder to price but directionally enormous - one locum month costs more than a GPU - and the coded-exhaust products (Charts) support quality reporting and coding efficiency that commercial scribes do not provide at all. Under the model's assumptions, hardware pays for itself in 12 to 20 months against the commercial-low scenario, and immediately against commercial-high. The honest caveats: the model assumes a hospital that can hire and retain two MLOps engineers, and Willow's benefits depend on the fine-tuning program actually being run - a license without a training plan is shelfware.

Implementation Roadmap

The roadmap is built around one principle: earn trust in a narrow lane first, then widen it. Phase 0 establishes the machinery of governance before any clinical touch; Phase 1 runs Willow Writer in two departments under silent-mode-plus-review; Phase 2 scales Writer, adds Charts and Sage with FHIR write-back; Phase 3 extends to Triage, Pulse and multi-site rollout with the EU AI Act conformity pack complete. Every phase ends in a go / no-go gate chaired by the steering committee with the kill criteria from Chapter 5 already in force. Table 9 lays out the windows, the work and the gates.

Phase	Window	Key activities	Gate to advance
Phase 0	Weeks 1-4	Governance charter; data-use agreements; de-ID gate online; synthetic (Synthea) end-to-end dry run; registry seeded	De-ID sampling QA at target recall; dry run clean
Phase 1	Days 30-90	Writer fine-tune on pilot-department notes; 2-department pilot; clinician review board; weekly audits	20%+ documentation-time reduction; zero PHI incidents; sign-off satisfaction 4/5
Phase 2	Months 4-9	Writer scale-out; Charts silent mode then threshold release; Sage guideline RAG live; FHIR write-back	Charts AUROC gate passed; Sage 100% citation audit; two clean audit cycles
Phase 3	Months 10-18	Triage with RN gate; Pulse digests; multi-site rollout; EU AI Act conformity pack finalized	Committee review; regulator-ready documentation accepted

Table 9. Phased implementation plan with gates

KPIs are tracked per product from day one - time-to-draft, edit distance, sign-off latency for Writer; extraction precision and AUROC for Charts; citation completeness and refusal quality for Sage - and every KPI has a pre-agreed threshold, because a metric without a threshold is trivia rather than

governance. The committee reviews the full KPI panel monthly, and the roadmap itself is versioned: delays are recorded with causes, which turns the implementation history into evidence of operational maturity rather than an embarrassment. Resourcing follows the same logic - Phase 0 requires roughly 0.5 FTE of legal and 1.0 FTE of informatics time; steady state is the three-person run team plus departmental clinical champions, whose role is to keep the workflow honest as scale grows.

Risk Register and Validation Protocol

Every residual risk worth managing is in Table 10, scored for likelihood and impact with a named mitigation and owner. The register is a living document in Sentinel - reviewed monthly by the steering committee, with additions welcome from any staff member through the incident channel. The most consequential risks are not exotic: hallucination entering the chart and PHI leakage at the gate are the two events that would end the program, which is why their mitigations (human sign-off on every output; de-identification plus sampling QA) are structural rather than procedural.

Risk	L	I	Mitigation	Owner
Hallucinated content enters record	Med	High	Citation grounding; clinician sign-off on 100% of outputs; weekly hallucination audit	CMIO
PHI leakage past de-ID gate	Low	Critical	Safe Harbor gate; 1% human double-read; zoned network; re-ID prohibition	Security officer
Model drift over time	Med	Med	Input/output distribution monitors; monthly eval suite; registry pinning	ML Ops lead
Upstream license change	Low	High	License-neutral architecture; weights pinned per version; legal review of derivatives	Legal
Shadow use outside workflow	Med	Med	SSO-gated apps; audit analytics for off-pattern usage; governance training	Clinical champion
Alert fatigue from Charts	Med	Med	Digest-first design; silent-mode thresholds; KPI-gated release	Ward leads
MLOps skill gap	Med	Med	Two dedicated FTE; vendor-neutral docs; runbook-first operations	CIO
Vendor/platform deprecation	Low	Low	Self-hosted weights; pinned containers; redeployable in one day	ML Ops lead

Table 10. Risk register (L = likelihood, I = impact)

10.1 Validation protocol

Validation is a permanent process, not a launch event, and it runs on four instruments. Benchmark gates: every model version passes a frozen internal benchmark suite (task-specific: MedQA-class for reasoners, NER F1 for extractors, AUROC for risk heads) before the registry will serve it. Silent-mode evaluation: every new or retrained model runs for one full quarter producing output nobody sees, while outcomes are collected and scored against the gate thresholds. Clinician review sampling: a fixed percentage of live outputs is double-read weekly by reviewing clinicians, with edit distance as the primary quality signal and findings reported to the committee. Red-teaming: a quarterly adversarial exercise in which clinicians deliberately try to induce unsafe outputs, with results scored and mitigated like any other defect.

10.2 Kill criteria

Each product carries pre-agreed kill criteria that trigger automatic suspension pending committee review: any PHI incident at the gate; hallucination audit rate above threshold in any weekly cycle; performance below the silent-mode gate; or two consecutive drift alarms on the serving layer. Suspension is a one-command operation per product - the router drops it from the serving pool - because a safety mechanism that takes an afternoon to execute is not a safety mechanism. Reinstatement requires a documented root cause, a revalidation pass and committee sign-off, and the full history is retained in Sentinel as part of the audit trail. The existence of credible, rehearsed kill criteria is, in the end, the strongest argument the hospital can make to itself and to any regulator that this platform is governed in fact and not only on paper.

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